

Discrete schedules and categorical representations

From time-indexed vectors to Catlab objects

This vignette shows how to apply interventions to discrete-time schedules and how to export program structure to Catlab objects.

```
using Catlab: nparts
using CategoricalInterventions
using Plots

default(; linewidth=2, markersize=4, grid=false)

beta = Target(
    :beta;
    kind=Parameter,
    value_type=Float64,
    algebra=MultiplicativeAlgebra(),
)

program = InterventionProgram(
    InterventionAtom(:closure, beta, Interval(2, 5), Scale(0.5)),
)

times = collect(0:6)
baseline = Dict(:beta => fill(0.30, length(times)))

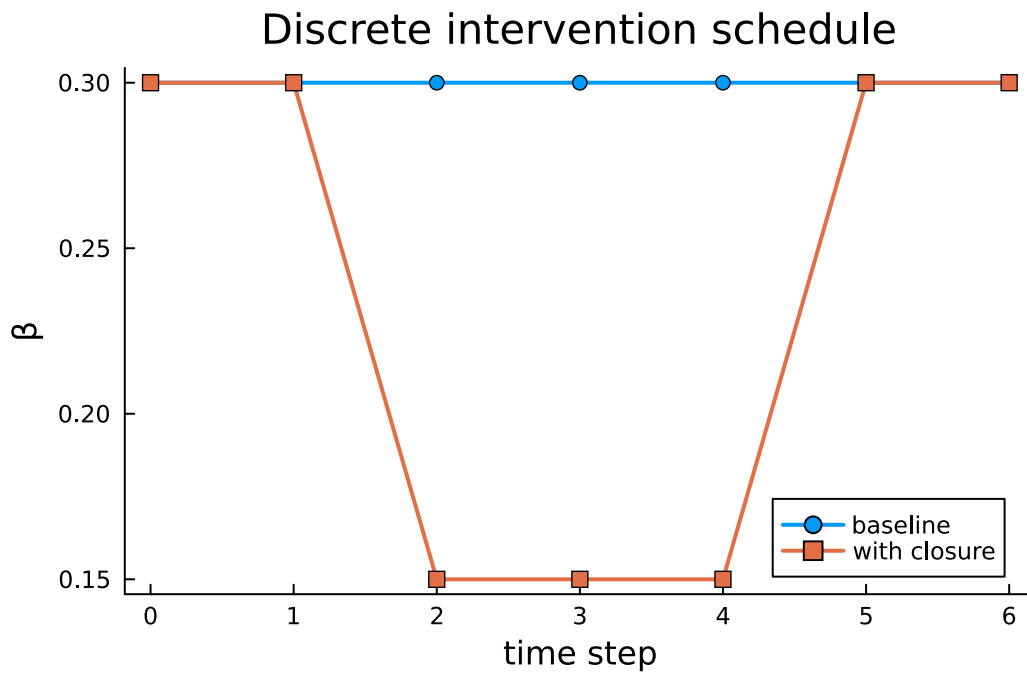
output = apply_discrete(program, baseline, times)
collect(zip(times, baseline[:beta], output[:beta]))
```

```
7-element Vector{Tuple{Int64, Float64, Float64}}:
 (0, 0.3, 0.3)
```

```
(1, 0.3, 0.3)
(2, 0.3, 0.15)
(3, 0.3, 0.15)
(4, 0.3, 0.15)
(5, 0.3, 0.3)
(6, 0.3, 0.3)
```

The intervention can be viewed as an edit to a time-indexed parameter schedule.

```
plot(
  times,
  baseline[:beta];
  label="baseline",
  marker=:circle,
  xlabel="time step",
  ylabel=" ",
  title="Discrete intervention schedule",
)
plot!(times, output[:beta]; label="with closure", marker=:square)
```



The same interval structure can be normalized into epochs and represented as a finite function from sampled times to epochs.

```

epochs = epochize(program)
active_times = collect(2:4)
f = epoch_finfuction(active_times, epochs)

collect(f)

```

```

3-element Vector{Int64}:
 1
 1
 1

```

epoch_finfuction expects times that lie inside active intervention epochs. For times outside those epochs, keep the original schedule values from apply_discrete.

The program can also be exported as an attributed C-set.

```

acset = as_acset(program)

(
  atoms = nparts(acset, :Atom),
  targets = nparts(acset, :TargetObj),
)

```

```

(atoms = 1, targets = 1)

```

The ACSet representation is useful when interventions should be treated as data: for example, for comparing programs, checking structure, or wiring interventions into larger categorical model-building workflows.